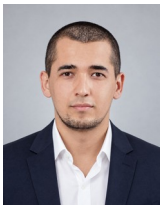
Sean Kakhkhorov



Applied AI & Backend Engineer (RAG · Python/FastAPI · TypeScript/NestJS)

Suwon, South Korea | kakhkhorov97@gmail.com | +82-10-7901-7555
Portfolio: https://www.sean-kakhkhorov.site | GitHub: https://github.com/kakhkhorov97
Work authorization: Bachelor's degree from a Korean university; eligible to change to an E-7 visa upon hiring

PROFESSIONAL SUMMARY

Applied AI and backend engineer with about 3 years of professional experience building web platforms, APIs, and Linux infrastructure for 20+ client organizations, plus RAG and LLM systems built end to end. Specialized in hybrid retrieval (BM25 + vector fusion via RRF, Korean morphological tokenization), retrieval benchmarking (Hit@K, Recall@K, MRR), citation grounding, and schema-enforced extraction. Bachelor's degree completed in Korean; fluent English.

TECHNICAL SKILLS

LLM and Guardrails: RAG, Gemini API (structured outputs, embeddings), citation grounding, abstention, Zod, Pydantic

Retrieval and Evaluation: Hybrid search (BM25 + vector, RRF), pgvector, ChromaDB, PostgreSQL FTS, Kiwi Korean tokenizer, Hit@K, Recall@K, MRR

Backend and Data: Python, FastAPI, TypeScript, Node.js, NestJS, GraphQL, REST, BullMQ, PostgreSQL, Drizzle ORM, MongoDB, Redis

Infrastructure and QA: Docker, Google Cloud Run, GitHub Actions (CI/CD, OIDC), Nginx, Ubuntu Linux, React, Next.js, pytest, Jest, Vitest (700+ tests)

PROFESSIONAL EXPERIENCE

IT Local — Full-Stack Web Developer | Tashkent · B2B web agency · full-time | Sep 2023 – Apr 2026 (contract ended)

- Core team of 4 developers (about 20 staff). Built an apartment-sales client's admin dashboard (Next.js, React/MUI, GraphQL, MongoDB) with 4 developers, owning **role-based access control (Super Admin / Manager / Sales Staff), the inventory GraphQL API and update mutations, and the admin table UI**. Also built and maintained websites for 20+ clients, including a construction client's systems.
- **Deploy automation:** replaced manual SSH deploys (15–20 min) with GitHub Actions CI/CD (4–5 min, push to live); started on my assigned client projects, then adopted by the team as a template on 5+ sites.
- **Servers:** grew from 5 to 15 Ubuntu servers, provisioning new environments (Nginx reverse proxy, SSL renewal, DNS, PM2, Docker, UFW/SSH hardening, automated backups).
- **Performance:** tens of thousands of catalogue records took about 3–5 s per search because of full collection scans; compound indexes and cursor-based pagination cut it to **under 1 s**.
- **Incidents:** traced an out-of-memory outage (about 15–20 min down) caused by uploads buffered in memory; added upload size/concurrency limits and PM2 auto-restart. Fixed 3–4 major outages; wrote API/deploy docs and a handover.

Golden Valley Investment (formerly Atlantis Investment) — Junior Backend Developer | Tashkent-based real-estate company · remote · project-based contract | Apr 2023 – Sep 2023

- Maintained the company's existing Express.js / MongoDB backend: bug fixes, small feature changes, JWT/session auth and file-upload related code, MongoDB indexes for search and pagination.

PROJECTS

Korea Visa Assistant — Bilingual Immigration Law RAG | FastAPI, Kiwi, ChromaDB, BM25, Gemini API, Cloud Run
Live: https://korea-visa-rag-jwq77bpivq-uc.a.run.app/ | **Code:** https://github.com/kakhkhorov97/korea-visa-rag

- Built an EN/KO RAG pipeline over 21 visa categories with statute-aware chunking, a Kiwi tokenizer that keeps visa codes (E-7-1, D-10) intact, and BM25 + Chroma RRF: 0.935 MRR, 96.9% Hit@5, 89.1% Recall@5 (vs. 0.702 MRR for BM25 alone, +33%).
- Added quote-grounding checks with raw-statute fallback, PII rejection, and SHA-256 verified index snapshots that remove startup embedding calls on Cloud Run.

Sofiya — Inventory-Grounded Hybrid RAG Concierge and Store | Next.js 15, NestJS 11, PostgreSQL (pgvector), Redis, Cloud Run
Live: https://sofiya.store/ | **Code:** https://github.com/kakhkhorov97/Sofiya-Bookstore/tree/v2

- Fused pgvector semantic search with full-text search via RRF; recommends only in-stock books with verified citations: 100% Hit@5, 93% Hit@1, 0.95 MRR (vs. 85% Hit@5 for keyword search, +15 points).
- Added idempotent Toss Payments checkout with server-side amount validation, Redis daily AI caps, and keyless OIDC CI/CD to Cloud Run.

CargoCheck AI — Trade Document Extraction and Verification | BullMQ, Redis, Gemini structured output, Zod, PostgreSQL
Live: https://cargocheck-ai-web.vercel.app/ | **Code:** https://github.com/kakhkhorov97/cargocheck-ai

- Cross-verifies invoices and packing lists with BullMQ workers and Zod-validated Gemini extraction, human-in-the-loop review, deterministic quantity/unit/decimal rules, and PDF audit reports.

CreatorHub and Lexara — Marketplace and Document RAG Backend | NestJS GraphQL, FastAPI, MongoDB, Socket.IO
CreatorHub: https://creatorhub.site/ | **Lexara:** https://lexara-top.vercel.app/

- Built an EN/KO/UZ marketplace with a Gemini assistant, idempotency keys, and 60-second lease-based notification workers; contributed FastAPI endpoints, uploads, and migrations to the Lexara document-RAG platform.

EDUCATION AND LANGUAGES

Chonnam National University (Yeosu, South Korea) — B.A. in International Trade, taught entirely in Korean | Sep 2019 – Aug 2023

Languages: Korean (completed a full bachelor's degree taught in Korean; can handle technical interviews and read technical documents), English (fluent), Uzbek (native), Russian (elementary)